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ORIGINAL PAPER

Transition from high school to university calculus: a study of connection

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Abstract
In Tunisia, calculus is a fundamental component of mathematics curriculum in high school and a major requirement at the advanced level in the mathematics bachelor's degree and in engineering. Yet, paving the way for a passage between high school calculus and university calculus remains a challenge for both mathematics education researchers and practitioners. In this paper, we address this issue by focusing on shared features between high school and university expectations in students' learning of calculus, and on the main impediments for high school to meet university requirements. We also propose an experimental method to help high school teachers to reconsider their actions according to insufficiencies in students' preparation for university calculus. The results show that it is possible to find a linkage between high school and university calculus so as to reduce differences and enhance students' transition.

Keywords Calculus · Transition · Limit · Calculus ideas · Connection

1 A search for continuity in calculus transition

When transitioning from secondary to tertiary education, students experience numerous kinds of changes. Researchers in mathematics education broadly acknowledge that differences in the way of understanding, doing and using mathematics stem not only from new learning expectations and the growth of cognitive requirements, but also from new teaching staff and practices, new classroom environments, increased autonomy and self-government, and a new social environment. Research in the calculus transition from high school to university has been addressed mainly through cognitive shifting needed to support the process of adjustment from prior understandings of calculus notions and procedural thinking to more advanced understandings and formal thinking (Bressoud et al. 2016). In a previous study, we investigated this issue via Tunisian textbooks and curricula of both levels (Bloch & Ghedamsi, 2005). Our analyses were conducted through a discontinuity hypothesis between high school calculus outcomes and university calculus requirements. This working hypothesis is generally accepted as a basis for research on transition that emphasizes differences rather than similarities. Our main results brought to light differences in learning expectations concerning proof, formalism, semiotic representations, and problem-solving tasks and technical methods. A complementary study on regular teaching actions in calculus courses at the beginning of university studies showed inefficiency in teacher management of differences and of students' moves towards university practices (Ghedamsi, 2015). Amongst the research results, the implication that teaching actions have an impact on students' learning has shed some light on crucial expectations related to informal and formal mathematics statements and mathematics reasoning that potentially could be bridged by high school calculus instruction.

Recently, claims for reducing the dualistic view of the transition—in terms of break and rupture—suggest that critical studies should illuminate opportunities to help students grasp differences and adapt to the changes efficiently (Gueudet et al. 2016). One pioneering vision is the search for a variety of continuous paths in the several changes faced by students entering university; a key proposal underlines a need to explore teachers' practices and teaching actions at both ends of the transition, and to examine disparities as well as possible connections between them. Research studies

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of this kind have mostly focused on teachers' pedagogical principles and characteristics on both sides of the transition, providing evidence of similarities and differences. For instance, Hong et al. (2009) and Wade et al. (2016) highlighted the necessity for high school and university teachers to communicate and to learn from each other despite their differences, since they both agree that high schoolteachers focus on students' needs, and university teachers give more attention to mathematics content. Regarding calculus transition, there are not many studies regarding relationships between high school and university teachers' practices and teaching actions that take into account specific cognitive concerns. The main exception is research on high school and university teachers' use of symbolic representations, which does not directly refer to calculus notions (Corriveau et al. 2017). This analysis showed important differences in the meaning of symbolism and the ways it is used at each teaching level. These meanings influence teachers' teaching actions (Thompson, 2016) and provide materials for strengthening the relationship between teaching actions on both sides of the transition.

Research on both sides of the calculus transition has also proposed some teaching plans to facilitate students' transitions. Among these interventionist studies, our research (Ghedamsi, 2008) has focused on fundamental definitions and theorems of existence related to number, limit and sequence taught at the start of university studies. Our findings revealed that students have difficulties with mathematical reasoning and the formal interpretation of informal calculus statements (e.g., on approaching infinity, numbers become progressively close). However, Haddad (2012) showed that it is possible to design tasks that allow students, at the end of study in Tunisian schools, to appreciate the connection between area, antiderivative and integral, and to ease the transition with Riemann sums as taught at the university. In the same spirit, Lecorre (2016) designed tasks for high school students to destabilize their prior understandings of fundamental calculus notions, and to make them feel the need to formalize their reasoning and progress in the elaboration of 'consensual' definitions (Ghedamsi & Lecorre, 2018). While most of the school interventions resulted in positive effects on students' learning of calculus, evidence for the positive impact of university interventions is not sufficiently convincing. However, at both levels, the implementation of the designed tasks in regular classes remains a challenging issue, mainly because the responsibility of the teacher in the elaboration of the teaching plans is underestimated. Clearly, high school teachers could have an important role to play in helping students with transition and adjusting their teaching actions' according to transition needs. Most of the research involving high school teachers has focused on teachers' knowledge and beliefs regarding teaching calculus. However, research focusing on the relationships between teachers' knowledge and teaching actions from the transition perspective are rare.

In this paper, we aim to add insights to the emerging trend of thought about transition by giving an 'optimistic' vision of calculus transition between secondary and tertiary education in Tunisia. For this goal, we needed to reconsider the perspectives that we had previously deployed, as follows: (1) to analyze textbooks and curricula (Bloch & Ghedamsi, 2005) in such a way that learning expectations, presented as considerably different, may be viewed as partly overlapping and not mutually exclusive; (2) to examine the calculus knowledge for teaching of university teachers (Smida & Ghedamsi, 2006) by focusing on knowledge related to students' learning of calculus and how this knowledge is met by high school calculus potential preparation. Then, according to what we learned from the two first steps regarding crucial learning expectations on both sides of the calculus transition, we put forward an experimental method to engage high school teachers in the project of building a potential continuous path between them. This method is based on collaborative work between high school mathematics teachers and mathematics education researchers. It has the following aims: (1) to model problematic teaching actions regarding students' predominant learning expectations in the transition; (2) to allow the teacher to experience confusion in these actions, to think on the meanings of calculus knowledge for teaching that have been implicit in these actions, and to carry out an experiment with a new meaning of the phenomenon, and with a change in his or her own knowledge.

Based on the data from three different studies concerning transition (one on mathematical content, one on university mathematics teachers' knowledge for teaching, and one on high school teachers teaching actions), our strategy is to reconsider the analysis of the data used in each study in such a way that the results permit us to address the possibility of making a linkage between the two sides of the transition.

2 Calculus trends in the transition[1]

2.1 Overview of the Tunisian context

In Tunisia, secondary education (Grades 10–13) is divided into two parts, one transitional year of common education and three years of pre-specialized education. It leads to the national baccalaureate diploma in mathematics, natural sciences, technology, economics and management, or

[1] For details about mathematics curricula and national textbooks see http://www.cnp.com.tn/french/index.htm and http://www.ipeit.rnu.tn/sites/default/files/Prog%20Maths%20MP%201%C3%A8re%20ann%C3%A9e.pdf.

humanities and social studies. The national mathematics curriculum of each grade level begins with core competencies for practicing mathematics followed by sets of specified domains, each domain containing knowledge and skills that students are expected to master by the end of the grade level. The competencies for practicing mathematics illustrate key mathematical proficiencies that students at all level should develop in order to understand, to do, and to use mathematics (e.g., to reason and to prove, and to communicate mathematics). Calculus is introduced for the first time in the 10thgrade. Depending on the specific content of the targeted grade level, it is organized through two major domains, as follows: functions including continuity, differentiation, integration, fundamental theorem of calculus and differential equations; and sequences including real numbers, while limits belong to both domains. Following secondary education, students who pass the baccalaureate examination can apply for medical schools, engineering schools, or bachelor's degree program. Engineering programs and bachelor's degree programs in mathematics require students to take courses in advanced calculus from the beginning of their studies. These courses are designed to provide 'a rigorous foundation for calculus', and usually cover the following topics: differential calculus including Taylor's theorem and differential equations; integral calculus including Riemann sum and the improper integral; infinite sequences and series including series of functions; and the real number system. According to the advanced calculus program, students are expected to demonstrate a high level of proficiency in advanced understanding of calculus notions and mathematical reasoning.

Secondary mathematics teachers are recruited among people who have a Bachelor's or Master's degree in mathematics and who pass theoretical and practical examinations set by the ministry of education. In grades 12 and 13, calculus classes are usually taught by teachers with considerable experience and the highest degree in mathematics. Almost all university teachers must hold a PhD degree to teach in Tunisian universities. To be recruited in public universities, they also must hold a teaching certificate and reach a designated score on tests of pedagogical knowledge.

In Tunisian secondary education, national mathematics textbooks—one textbook for each grade level—represent the implemented curricula and the fundamental resource that mathematics teachers must use in classrooms. Homework and assessment tasks assigned to students are usually more or less similar to those that students actually encounter in classroom. Thus, the investigation of the high school intended curriculum and the national textbooks provides us with appropriate information concerning what students are expected to be learning. At the tertiary level, mathematics courses are usually organized into lectures—formal talks delivered to set definitions and theorems, and tutorials—settings to apply definitions and theorems already studied, and to help students enter more deeply into mathematics thinking processes. These courses can be given either by the same teacher or shared by two teachers.

In what follows, we present the method we used to investigate the common features between the two sets of expectations in the learning of calculus in the transition, namely, those manifested in both high school and university curricula and instruction.

2.2 Method

The majority of research on students' learning of calculus and cognitive requirements in the transition has been conducted from the following perspectives: practical demands vs. analytical and theoretical demands; computation and visualisation vs. formalization; instantiation vs. generalization; conception and image vs. definition, and calculus notion viewed as a process vs. calculus notion encapsulated in an object (Bressoud et al. 2016). Thus, a feature of the trends of research on calculus textbooks and curricula is that no global exhaustive framework is assumed because there are too many didactical variables to consider. In Brousseau's (1997) theory of didactic situations, didactic variables are defined as parameters that influence the way students may solve a task and can be used to describe the cognitive demands of tasks, such as the use of technical methods, the use of visualisation techniques, the use of formalism, and the use of reasoning and proof setting. In our previous study of curricula and textbooks regarding limit of function and limit of sequence (Bloch et al. 2005), we deployed a set of these variables to deepen insights on differences, by examining the same data from several separate perspectives. In this paper, we are not focusing on the variables themselves but on relationships among them that can enable the description of possible continuous paths between calculus tasks of high school and advanced calculus tasks.

Curricula of both sides highlight students' conceptual understanding of calculus notions as a cornerstone goal of teaching calculus, and explain several skills intended to shape this understanding. These skills are fundamentally concerned with mathematical reasoning, the association of essential calculus ideas, and the knowing of each idea through given definitions and theorems. The study of the didactic variables concerned with the learning expectations related to this teaching goal permits us to outline the nature of conceptual understanding as conceived at each level. In Tunisia, the limit notion is a cornerstone notion of calculus instruction; it unifies calculus practices in the Archimedean continuum and connects almost all foundational calculus ideas that are taught on both sides of the transition, including the following: real numbers as a study of limits of sequences; differential calculus as a study of limits of rates of change;

Table 1 Calculus chapters as introduced in chronological order

12th grade	13th grade	1st year of university
Algebraic properties of functions	Continuity and limit	Generalities on fundamental techniques
Continuity	Sequences' convergence (8 T)	Real numbers and sequences (3 T)
Continuity and limit	Derivatives (7 T)	Limit, continuity, differentiability (4 T)
Asymptotic behaviours	Inverse functions	Convex functions
Derivative numbers (3 T)	Antiderivatives	Asymptotic behaviours
Derivative of functions	Integral (3 T)	Integration (includes antiderivatives) (1 T)
Calculus of algebraic functions	Logarithmic function	Series
Trigonometric functions	Exponential function	
Sequences	Differential equations	
Sequences' convergence (6 T)		

integral calculus as a study of limits of accumulated changes; and the fundamental theorem of calculus (FTC) as a study of the relationships between two limits. In turn, usefulness of the unifying idea of limit depends upon three factors, as follows: (1) a precise definition of limit; (2) in each application, a proof of existence of this limit, and (3) a method for 'calculating' the limit. Unlike algebra formulas of limit, successive approximation methods put forward the use of infinity and infinitesimals, stress the difference between algebra and calculus, and build bridges with fundamental structural constructs. Reports on the epistemological and cognitive basis of this claim are numerous and shed light on important issues that we cannot discuss here (Monaghan, et al. 2019). In Tunisia, real numbers are introduced for the first time in the algebra courses of the 9th grade and the step of going to the limit is made in the 12th grade. Several examples of algebraic functions are introduced symbolically and graphically during the 10th and 11th grades, and broadly defined as rules that assign one member of the range to each member of the domain; the algebraic idea of rate of change—how f(x) change as x change and reciprocally—appears to be absent. Accordingly, in this paper we focus on definitions, theorems, and tasks given in the two last grades of school and the first year of the university regarding the conceptual understanding of limit according to the following aspects: (1) reasoning with the limit notion, including inductive reasoning, counterexamples, reductio ad absurdum, necessary condition, or sufficient condition; (2) linking the limit idea to real numbers, derivatives, integrals, or the FTC. We considered students' learning expectations from the relationships among three variables, taken from among those that we have previously deployed. For tasks, we analyzed semiotic representations of limits (e.g., numeric, algebraic, graphic, and formal representation), the limit status (process, procept, object), and reasoning and proof setting (e.g., algebraic formulas, numerical approach through inductive reasoning, graphical representations and counterexamples, formal proof and deductive reasoning). Definitions and theorems were analyzed by means of the representations and proofs that were involved. Three main trends could be identified in this analysis, as follows: (1) one concerning students' images of limit; (2) one focusing on both students' images and structural elements involving limit; (3) one growing through deductive constructs based on formal definitions and proofs. In addition to high school textbooks, we collected a set of advanced calculus courses—given for students in engineering—composed of lecture notes and all tasks assigned to students in tutorials. We chose to focus on calculus texts involving limit among three topics, namely, real numbers and sequences, derivatives, and integrals. For both sides, the sample included (1) introductory tasks for defining sequence convergence, derivative number, and integral, (2) introductory tasks for setting theorems of existence of mainstream notions related to sequence convergence, derivative number and integral (e.g., intermediate value of monotonic function, mean value for derivative, mean value for integral), and (3) all application tasks for investigating the numerical existence of limits through approximation procedures in such a way that these definitions and theorems become the arguments to validate the used procedure. Table 1 shows the allocation of these tasks in calculus chapters (3 T means three tasks).

Below, we demonstrate our findings on calculus trends in the transition, and possible impediments for high school students in meeting university requirements.

2.3 A search for a potential passage

We carried out three studies of high school and university calculus texts, one for each chapter, namely, real numbers and limits (C1), derivatives and limits (C2), and integrals and limits (C3). The units of analysis were tasks regardless of the number of questions. In line with the university curriculum, these chapters adopt a theoretical-driven approach to introducing calculus notions. We noted that definitions and theorems are usually followed by examples and counterexamples. Although high school curricula and textbooks

Table 2 Organization of calculus tasks, definitions and theorems from high school to university

	Grade 12	Grade 13	1st year university
C1	D1, Th1.1, Th1.2 TD1, TTh1.2 TA1.1 (1) TA1.3 (2) TA1.4 (1)	Th1.3, Th1.4, Th1.5, Th1.6 TTh1.3, TTh1.4, TTh1.6 TA1.1 (2) TA1.2 (2) TA1.3 (1)	TA1.2 (2)
C2	D2.1, D2.2, D2.3 TD2.1, TD2.2, TD2.3	Th2.1, Th2.2, Th2.3 TTh2.2, TTh2.3 TA1.2 (1) TA2.2 (2) TA2.3 (2)	TA2.2 (1) TA2.3 (3)
C3		DS3 (FTC), Th3.1, Th3.2 TDS3 TA3.1 (1) TA3.2 (1)	DU3 TA3.2 (1)

emphasize the use of examples and contextualized tasks, the definitions of convergence of sequence, derivative number, and integral are enunciated precisely, and theorems are formally proved after they are exemplified. Table 2 shows the organization of tasks, definitions and theorems (TThi.j refers to introductory task of Thi.j theorem; the other codes are described below).

In the university, C1 is founded on the completeness axiom of the set of real numbers, 'any non-empty subset bounded from above has a least upper bound' and aims to establish rigorously the properties of real numbers and their interpretations by means of sequences. The curriculum recommends highlighting the role of sequences in approximating real numbers. In school, the link between real numbers and sequences is stated through tasks involving successive approximations. The theorems of limit of geometric sequences (Th1.2), necessary and sufficient condition of convergence using odd and even subsequence (Th1.3), sufficient condition using bounded sequence (Th1.4), and adjacent theorem sequence (Th.1.6) are not admitted, contrary to the theorem of uniqueness of limit (Th1.1), and the monotonic and bounded theorem (Th1.5). All these theorems are preceded by an introductory task except Th1.1 and Th1.5. The definition of sequence convergence (D1) is enunciated formally after being introduced through an approximation sequence of an irrational number that is algebraically and graphically represented. The task is about finding n given successive approximations of the number with fixed errors, using a typical example of an increasing and bounded sequence, and the task needs algebraic skills concerning inequalities. As a result, the task excludes infinity and infinitesimals and focuses on the algebraic inequalities in the formal definition. This focus leads to a poor image of limit or to a definition simply learnt in rote fashion for the needs of algebraic manipulations of inequalities (Ghedamsi et al. 2018). Similarly to university practice, the definition is used to prove theorems using algebraic inequalities and epsilon statements. The analysis of application tasks on both sides of the transition permitted us to classify them into four types, as follows: approximation of a rational number given by means of its improper representation (TA1.1); approximation of an irrational number (TA.1.2); approximation of polygonal area, perimeter and length (TA1.3); approximation of volume of circular solids (TA1.4). Depending on the task, procedures of approximation are theoretically founded on Th1.2, Th1.3, Th1.4, or Th1.5.

In the university, the curriculum related to derivatives (C2) recommends the use of graphic interpretations as much as possible, and strengthens the use of Rolle's theorem (Th2.3) to prove the existence of a fixed point of the k-Lipschitz function, and the mean value theorem for derivatives (Th2.2) to approximate this point. In school curricula, the link between approximation methods and foundational notions related to derivative is not clearly stated. While Th2.2 and Th2.3 are enunciated graphically and formally before being proved and after being instantiated, the extreme values theorem (Th2.1) is simply admitted. The first introductory task (TD2.1) is about a parabolic model of velocity and the question is to find the instantaneous velocity defined as a limit of the average velocity between $a+h$ and a. The second task (TD2.2) introduces the tangent line at a known point as the line that has a slope equal to the derivative at that point. The data consists of algebraic and graphical representations of a polynomial function, a secant line through a known point A and an unknown point M of the graphic. The question is to compute the slope of the secant and to conjecture the equation of the secant (AM) when M approach A in such a way that the secant line turns into what is called the tangent line when M approach A. Unlike previous tasks, this task employs statements linked to the approaching processes that are not precisely defined. An introductory task follows, of the linear approximation of a function (TD2.3) founded on the idea that it is easier to use the tangent than the curve to find an approximation of $f(a+h)$ when h approaches zero.

Table 3 Limit status and representations among the tasks

	Limit representation	Limit status						
		Numeric	Graphic	Algebraic	Formal	Process	Procept	Object
12th	C1	3	4	6		2	4 ATi.j	
	C2	3	4	8			5 ATi.j	
	C3		1	1	1		1ATi.j	1 ATi.j
13th	C1	3	3	3		3		
	C2	3	3	7			5 ATi.j	2
	C3	1	1	2	2		1 ATi.j	3 ATi.j
1st	C2	1	2	3			2 ATi.j	1
	C3			1	2			1 + 1ATi.j

The aim of these tasks is to show that (1) the way to find a more accurate instantaneous velocity is to reduce the interval of time used in the computation, and (2) the slope between points of the distance curve that are closer and closer on the curve is equal to the velocity at a single point. Additional interventions of the teacher should make the link clear and help students connect approaching processes and algebraic statements. Although we agree that there is an implicit intention to give a meaning to limit through derivative at a point, and contrary to university, to go beyond algebraic manipulation of inequalities, the high school teacher has the responsibility of managing students' formulations of expressions involving approaching processes. In light of analysis of the task in both levels, two kinds of application tasks theoretically founded on Th2.1, Th2.2, and Th2.2 are added, namely, approximation of zero of algebraic functions (TA2.1), and approximation of zero of transcendental functions (TA2.2). In school, there is no preliminary algebraic and graphical work with rates of change before introducing derivative number.

In the university, C3 is founded on the construction of the integral using Riemann sums. Definite integral (DU3) introduced through Riemann integral is accompanied by a geometrical interpretation of the limit of Riemann sums as the area under a curve, if that limit exists. This content is followed by other topics covering mean value theorem for integrals (Th3.1), antiderivatives and indefinite integral (Th3.2), and FTC. In school, the existence of antiderivative of continuous function is admitted and introduced before integral through a small chapter containing algebraic formulas and some initial computational methods. Definite integral (DS3) is firstly established as the area under the curve of a continuous positive function which is in turn defined similarly to FTC, the definitions of the definite integral of continuous function, and the area under the curve of continuous function are then stated accordingly. Th3.1 and Th3.2 are successively introduced and proved using these definitions without introductory tasks. The introductory task for definite integral (TDS3) is based on the algebraic and graphical representations of a parabolic function f as well as the graphical representation of what is called area under the curve of f between t and 1 ($t > 1$)and noted $A(t)$ (the given t is taken on the interval where f is monotone). The task contains questions for finding inequalities that involve $A(t+h)$-$A(t)$ as well as $A(t)$–$A(t-h)$ in such a way that the use of algebraic rules of limit and properties of antiderivative leads to $A_g'(t)=f(t)$ and $A_d'(t)=f(t)$ so $A(t)=F(t)–F(1)$, where F is an antiderivative of f. As stressed by Haddad (2012), this task dissimulates the indefinite integral, and creates an ambiguous link between area, antiderivative and definite integral. Mathematics education research highlights three fundamental issues for learning, integral, and FTC: (1) the parallels between sums and integrals, and between differences and derivatives; (2) the link between integral (accumulation of total change) and derivative (rate of change); (3) the role of successive approximations of areas for establishing this link. Although students' images of integral were already weak before entering university (Haddad, 2013), we did not observe a specific focus on integral issues in university texts. In both cases, students are required to 'know' definitions and theorems related to integral, and to perform techniques to use them. Other kinds of application tasks complement the set of tasks previously identified, including the following: approximation of a real number represented by successive terms of a series which is described at the high school as a particular sequence (TA3.1); approximation of non polygonal area (TA3.2). The theoretical arguments involve Th3.1, Th3.2 and FTC. Table 3 describes limit representation and limit status for the whole tasks (multiple representations of limit may be involved in the same task).

Unlike the case of university, in all the approximation tasks of school, limit operates dually as process and object, thus as a procept (Tall et al. 1999) that brings together deep meaning and techniques. Almost all the kinds of representations of limit are involved in these tasks and require students to make connections between them. At the university, the structural approach remains dominant even when the tasks are supposed to underline intuitive ideas of the limit and

promote approximate computations. Although introductory tasks of school lack consistency from the cognitive perspective, the requirement in term of algebraic manipulations of inequalities probably meets university expectations. A closer look at reasoning and proof settings revealed the apparent dominance of formal proof and deductive reasoning (e.g., necessary and sufficient conditions, reductio ad absurdum) in the university. This finding is not surprising, yet we expected that approximation tasks, when efficiently organized, would endorse the use of more heuristic tasks. We also noted that approximation tasks in school allow more opportunities for connecting graphical arguments to numerical computations. The manipulation of algebraic inequalities constitutes the interface between the former practices and the final argument to validate answers.

Both sides of the transition focus on algebraic manipulation of inequalities and leave a small place for the emergence of limit ideas through intuitive and experimental manipulations. For both sides of the transition, understanding calculus ideas and their connections means developing a network of computational knowledge with algebraic inequalities in which limit ties these ideas and techniques together. Even if we have much to say about these choices, we do not add more discussion of the cognitive output of students; this is not the aim of this paper. Despite the convergence of points of view between high school calculus and advanced calculus program, there are at least two fundamental issues that may damage the passage between the two sides, and which should be revised in school: (1) high school calculus textbooks do not sufficiently focus on mathematical reasoning rules; and (2) high school calculus textbooks do not highlight the link betweeninformal statements about limit involving imprecise expressions (e.g., 'approaching more and more closely') and definitions.

3 University teachers' knowledge of newly admitted students

A complementary study of university teachers' knowledge for teaching calculus has the aim of advancing understanding of the main impediments for school calculus to meet university requirements.

3.1 Background

University teachers' interpretations of the teaching goal related to conceptual understanding may be firstly appreciated through the analysis of lecture notes and tasks assigned to students. We understand these interpretations as part of teachers' calculus knowledge for teaching 'CaKT' (Ball et al. 2008), which are structured through four domains adapted to calculus, as follows: (1) Common Calculus Knowledge 'CCaK' includes scholarly-related knowledge used by university teachers as mathematicians, such as knowledge of topology; (2) Specialized Calculus Knowledge 'SCaK' consists of calculus knowledge used by teachers to prepare teaching plans, such as knowledge needed to define calculus notions, knowledge of paradigmatic examples, and knowledge of high school calculus; (3) Knowledge of Calculus and Teaching 'KCaT' comprises knowledge used in a given calculus teaching situation, such as knowledge needed for deciding on the use of specific proofs, for explaining limit ideas (for instance by using different representations), and knowledge used for emphasizing relationships between limit ideas; (4) Knowledge of Calculus and Students 'KCaS' includes knowledge of students' difficulties with calculus notions, such as knowledge of students' inappropriate ways of reasoning with calculus notions, knowledge of students' conceptions of limit notion, and knowledge of possible enhancements of high school students' knowledge to meet the requirements of university calculus.

In our previous research aiming to complement lecture notes with task analysis (Smida et al. 2006) we collected data from a questionnaire completed by 57 university teachers (21 female and 36 male) teaching calculus in the first year of the university. Although our sample was not randomly constructed, it was quasi-representative of calculus university teachers from 4 Tunisian universities (1 in the north, 2 in the centre, and 1in the south). At the time of the survey, 13 teachers had less than eight years of teaching experience and 14 teachers had more than 16sixteen years of teaching experience. While we assumed that the questionnaire should contain concrete calculus tasks, it was difficult to ensure an experimental setting that engaged university teachers in those tasks. The questionnaire involved, among others, multiple-choice and multiple-response questions for collecting information on KCaT and KCaS. In this paper, we focus on a subset of 11 questions (from a total of 31) dealing with two domains of teachers' knowledge, KCaT and KCaS, including explanation of relationships between calculus ideas ($KCaT_1$), what particular tasks should be emphasized to overcome students' difficulties ($KCaS_1$), and what calculus knowledge encountered in school should be improved during high school ($KCaS_2$). Even if, ultimately, we collected declarative sentences from the teachers and little evidence of their knowledge could be derived from the data, we aimed to advance understanding of calculus learning needs at the entrance to university courses, and to identify those that could be managed by high school teachers through viable changes in their actual teaching actions. We used three multiple-choice questions to study $KCaT_1$, namely, use of graphics to support the introduction of definitions (Q12), use of examples when enunciating theorems (Q11), and use of formalism to define notions (Q10). We used seven

Table 4 Percentages of teachers and means for $KCaT_1$ items

	Use Formalism Q10	Use examples Q11	Use graphics Q12		Q10	Q11	Q12
Never	8.77	7.02	7.02	Mean	2.89	3.27	2.89
Sometimes	28.07	12.28	22.81	Std.Deviation	1.03	0.94	0.89
Half time	21.05	26.32	42.11	Min	1	1	1
Almost all	35.09	52.63	26.32	Max	4	4	4
Total	92.98	98.25	100.00	Valid	53	56	56
Missed	7.02	1.75	1.75	Missed	4	1	1

Table 5 Percentages of teachers for $KCaS_1$ items

	Procedures Q1	Logic rules Q3	Graphics Q4	Symbolic calculator Q5	Graphing calculator Q6	Heuristic Q7	Conjecture Q8
Never	5.26	3.51	1.75	75.44	91.23	36.84	40.35
Sometimes	43.86	29.82	38.60	15.79	1.75	38.60	40.35
Half time	35.09	42.11	40.35	5.26	3.51	19.30	10.53
Almost all	15.79	24.56	17.54	3.51	1.75	5.26	5.26
Total	100	100	100	100	98.25	100	96.49
Missed	0	0	1,75	0	1,75	0	3,51

Table 6 Means for $KCaS_1$ items

	Q1	Q3	Q4	Q5	Q6	Q7	Q8
Mean	2.61	2.88	2.75	1.37	1.14	1.93	1.8
Std.Deviation	0.82	0.83	0.77	0.75	0.55	0.88	0.85
Min	1	1	1	1	1	1	1
Max	4	4	4	4	4	4	4
Valid	57	57	56	57	56	57	55
Missed	0	0	1	0	1	0	2

questions to study $KCaS_1$, namely, procedural tasks (Q1), tasks involving rules of mathematical logic (Q3), graphical tasks (Q4), tasks promoting the use of the symbolic calculator (Q5), tasks promoting the use of the graphing calculator (Q6), heuristic tasks (Q7), and tasks in which students are expected to make conjectures (Q8). Each of these questions had four possible responses, namely, 'never', 'sometimes', 'about half the time', and 'every or almost every time'; the responses were successively coded from 1 to 4. We also compared the means of teachers' responses to different items (e.g. graphical tasks (Q4) and heuristic tasks (Q7)), by means of a paired t-test that was used to find statistically significant differences. An additional 4-response question (Q16) was included to gain information about $KCaS_2$. The responses were numbered regarding each teacher's most convenient choice and coded accordingly. The responses dealt with formal statements (Q16a), rules of mathematical logic (Q16b), reasoning (Q16c), and understanding notions (Q16d). In the following section, we present our findings on university teachers' knowledge for teaching calculus in the transition.

3.2 On calculus knowledge for teaching in the transition

In Q21 of the teachers' questionnaire, we asked the teachers if they plan their teaching according to the program in terms of chronology and content. More than 75% of the teachers declared that they followed the program and almost all the recommendations. This result gives more credibility to our findings concerning university calculus content. Teachers' responses to the questionnaire allowed additional insight concerning how they teach this content (Table 4). Following descriptive information concerning the occurrence of items and the percentage of teachers reporting those items, we present statistical results regarding associations and differences among the items. Tables 5 and 6 show the percentages of teachers and means according to $KCaT_1$ items. More than 90% of the teachers

reported that they do not give tasks that require the use of the graphing calculator. In Tunisia, social factors may sometimes hinder the use of technology in education. For this reason we did not include this item in the remainder of the analysis.

Table 4 gives percentages of teachers and means according to $KCaT_1$items. About 9% of teachers declared that they were not concerned about the use of formalism for defining calculus notions while more than 55% of them employed formal statements at least about half the time. These proportions are not surprising; university teachers give a particular role to formal statements in explaining definitions, but they may also add graphics and examples: more than 65% of teachers affirmed that they use examples, and the same percentage holds for the use of graphics, at least about half the time. Yet, this result has no valuable meaning from the transition perspective as we have no information about how teachers explain the connections between graphic and formal representations through examples. We assumed that the nature of tasks that teachers assign to students for overcoming difficulties may give information about their calculus knowledge of cognitive demands of tasks ($KCaS_1$). In this study, the nature of these tasks was stated according to calculus learning requirements and expressed through different items. More than 70% declared that they never ask students to use symbolic calculators. This result corroborates our finding about the apparent absence of numerical representation of limit in university calculus. About 35% of teachers declared that they do not assign heuristic tasks to students, while about 40% of them claimed that they do not ask students to make conjectures. From a teachers' knowledge perspective, the same interpretation may be given for each result, for instance: (1) these teachers know that such tasks may engage students in intuitive considerations of limit, so leading to potential confusion that it is important to avoid; (2) experimentations, or conjectures, do not inform students' thinking concerning calculus notions; (3) these teachers are not acculturated to these practices and do not have calculus knowledge that is appropriate for such usage. More than 65% of teachers declared that they assign tasks to improve students' manipulation of logical rules at least about half the time. This aspect could be of concern when considering transition issues, since it may imply that teachers know that students enter the university with a lack of knowledge of logical rules. It also may be interpreted simply that teachers know that students should manipulate more formal statements to meet program expectations. Nearly half the teachers reported that they give students tasks to perform their learning of procedures at least half of the time, which is an expected result, while almost the same proportion said that they assign graphical tasks at least half of the time. Depending on the relationship between the two items, when we focus on teachers' knowledge the results may be interpreted as teachers knowing that learning procedures should involve graphics to facilitate students' interpretation of fundamental steps and to avoid maladjusted understanding.

Following paired t-tests, we were able to make preliminary conjectures on $KCaT_1$ and $KCaS_1$. When testing $KCaT_1$ items, we found that there was no significant difference between the means of teachers' responses to the use of formalism and the use of graphics. The contingency table of our sample showed that more than 60% of teachers deploy formal statements for explaining definitions and support their explanations by using graphics. However, there were no significant differences between the means of teachers' responses to the use of examples and the use of graphics, and to the use of examples and the use of formal statements. According to the contingency table, this same percentage of teachers used examples to illustrate theorems involving fundamental calculus notions. Contrary to university texts, calculus knowledge for teaching may contain additional elements related to graphics and examples that are used to explain ideas. The testing of the relationships between $KCaS_1$ items showed that there was no significant difference between the means of teachers' responses to experimental tasks and to tasks promoting conjectures. From the transition perspective, for more than 60% of the teachers', KCaS includes tasks that may form a bridge with high school calculus tasks. Additionally, there were no significant differences between procedural tasks and tasks that involved logical rules, between procedural tasks and graphical tasks, and between tasks that involved logical rules and graphical tasks. From the transition perspective, for more than 60% of the teachers, $KCaS_1$ may include knowledge that has the purpose of helping students' manipulations of logical rules. A complementary look at $KCaS_2$ items showed that about half the teachers classified mathematical reasoning as the most crucial learning need for newcomers, arguing that secondary education should give more attention to this insufficiency.

This preliminary investigation of university teachers' CaKT revealed the following three main observations: (1) although, calculus knowledge for teaching consists mostly of formal statements, additional knowledge involving graphics may be deployed to illustrate definitions; (2) similarly to school textbooks, teachers' calculus knowledge concerning students does not involve heuristic tasks; and (3) calculus knowledge concerning students includes students' shortcomings university teachers declared that they enhanced students' manipulation of formal statements, and devolved to school the responsibility for overcoming reasoning issues.

4 High school teaching actions beyond school calculus

In this section we briefly outline an empirical approach aimed at helping high school teachers adjust their teaching actions in accordance with university expectations; and we give some first comments on its use to promote a linkage between high school and university calculus, so as to reduce differences and enhance students' transition.

4.1 Theoretical background

It is acknowledged that teachers' CaKT influences teaching actions, yet the relationship remains unclear. However, while teaching, teachers need to be aware of students' learning needs and to adjust their actions accordingly. This adjustment necessitates that teachers have substantial CaKT to manage these demands. The value of reflection on teaching actions in order to enhance teaching has been recognized for decades (Dewey 1933; Schön 1983); however, it remains difficult to apply in practice. We assume that changes in a teacher's teaching actions require changes in that teacher's CaKT. In turn, the quality of changes in the teacher's CaKT depends on her/his ability to reflect on the meanings she/he gives to this knowledge (Thompson 2016). This kind of reflection is a well-established concept known as reflective thinking. Reflective thinking encompasses the capacity for conscious considerations of the consequences of the teaching actions on the progress of students' learning, and deep examination of the causes embodied in the related CaKT (Larrivee 2000). Reflective thinking requires the examination of at least three kinds of content of the teacher's reflection, as follows: (1) technical, related to teaching context including that of the school and programs; (2) practical, linked to the experience as both teacher and graduate of mathematics; (3) theoretical, including general assumptions and rules. However, the capacity to engage in reflective thinking is initiated only after the recognition of problematic class situations and uncertainty; reflective analysis often requires collaborative effort. One of the mechanisms for reflective thinking involving collaboration between teachers and researchers is what is called guided reflection (Husu et al. 2008). Guided reflection begins with recording and transcribing particular lessons, while questions for stimulated interviews are constructed by the researcher according to the observed problematic situations. The first aim of the interview is to allow the teacher to bring to mind the actual classroom situation as clearly as possible, to be confronted with a dilemma, and to verbalise thoughts in connection with the actions. The teacher describes consequences of her/his actions, and provides possible causes. The second aim is to revisit the teacher's CaKT and explore the possibilities for changing actions. In our previous study (Ghedamsi et al. 2018), we identified the teaching actions of an experienced teacher in a Tunisian school by directly observing six successive calculus classes on sequence and sequence convergence (each class lasted two hours and involved 25 high school students). As revealed by this research, these teaching actions had an impact on learning calculus in the transition (Robert et al. 2001). In the following section, we restrict our focus to some problematic situations and related teaching actions that we have used to support teachers' reflections on their impact upon students' advanced understanding of calculus. In particular, we present some preliminary results founded on an experimental method of guided reflection, designed to push mathematics teachers beyond superficial interpretations of classroom issues, and which facilitates reflective thinking on actions. We made an attempt to go beyond the status quo of teaching calculus in school by engaging high school teachers in reforming their actions and building a passage for students towards university calculus.

4.2 Teaching actions in high school calculus classes: possibilities to support the passage?

The study reported in this section forms part of a research project intended to make mathematics education research relevant to high school teachers. Building on the theory of Brousseau, we elaborated a methodological tool for lesson analysis founded on the interactions taking place within the system of relationships between teachers' mathematical knowledge for teaching, students' prior knowledge, students' informal understandings, and mathematical knowledge—including preconstructed tasks, tools, graphs, symbols, etc. (Ghedamsi et al. 2018). The interactions were enacted according to the role of both students and teacher during three learning phases, namely, a phase of students' action on mathematics objects, a phase of students' oral and written formulations about mathematics objects, and a phase of students' validation. The application of the tool for analyzing classroom situations regarding sequence and sequence convergence allowed findings related to teachers' actions through several classroom situations. Among these actions, we focused on those that demonstrated the teacher's unawareness of transition issues, and which are related to the management of the following: (1) students' use of informal statements; (2) students' use of the rules for use of quantifiers; (3) students' reasoning through the use of necessary and sufficient conditions; and counterexamples. Our project was designed to engage the teacher in a collaborative process, based on guided reflection, through a series of interview sessions planned to make him aware of those teaching actions, to reflect on the related knowledge and to engage in efficient change.

Table 7 Examples of classroom sequences

Informal statements	Quantifiers' rules
P: what do you think about this sequence? *S1: its terms are approaching √3* *P: and…* *S2: they are close to √3 and never reach it* *P: it isn't clear with the graphic… let's move to another tool…use your calculator* *S3: terms are approaching more and more* *S4: and never reach it!*	*S2 (blackboard): f is bounded from above then for every* $x \in D_f$ *there exists M such that* $f(x) \leq M$ *P: look at your statement… should you find M for each x?* *S5: No!* *P: then M exists…* *S2: for every x* *S2 (blackboard): [correction of statement]* *Students all together: what's the difference?*

For preparing the interview sessions, the most difficult task that we faced was the a priori evaluation of teacher's calculus knowledge for teaching with regard to classroom situations and teaching actions. Yet, we were able to conjecture a lack of knowledge concerning KCaS and KCaT and to elaborate materials for guiding the teacher's reflection. For instance, the teacher's knowledge set did not include knowledge of relevant tasks about quantifiers' rules and the connection between informal and formal statements. Table 7 briefly exemplifies two sequences from classroom situations concerned with informal statements and quantifiers' rules (utterances were translated verbatim from French; P refers to teacher and S to students).

In addition to classroom transcripts, these materials were used during interview sessions to help the teacher think about the meaning of the knowledge involved in his actions, reconsidering this knowledge if necessary, and exploring other alternatives. To provide a basis for discussion, we asked the teacher several types of questions, for example the following: Is there anything that strikes you in this classroom sequence? (KQ1); What kinds of calculus notion and understanding does this classroom sequence involve? (KQ2); Why did you do [this] at this point? (KQ3); What might have happened, if you have done the same before? (KQ4); How can you improve what is not working? (KQ5). We briefly report on some data and preliminary results of the analysis of four short sequences from the first interview session related to formal statements, quantifiers' rules, and reasoning. The classification of utterances is presented in Table 8.

To answer the question of what constitutes evidence of reflective thinking in teacher's utterances, we mostly focused on the kinds of content of the teacher's reflections regarding his actions. Table 9 provides some typical teacher's responses to the aforementioned types of questions (each utterance, translated verbatim into English, is preceded by the code of the appropriate type of questions).

With regard to the possibility of supporting teachers' changing knowledge and adapting teaching actions, the results showed that guided reflection involving collaboration with a mathematics education researcher may be a lever for regulating a teacher's decision-making processes. Although we have been able to identify crucial aspects that are essential to initiate adaptation of the participating teacher's knowledge, more interviews are needed to improve our understanding of the meaning given by the teacher to this knowledge. From the transition perspective, the teacher's specialized calculus knowledge (SCK) is appropriate and could help him build up his repertoire ofKCaT and KCaS (including use of examples and counterexamples, informal statements and topological meanings, instantiation of quantified statements). Even if these potential changes fundamentally depend upon the expectations of the school program, they engage a high school teacher in making the effort to go beyond these expectations. This collaborative method is a time intensive process for both teacher and researcher, but it can be a satisfying experience for both. In the context of the participating teacher, one benefit for teaching that we discovered is that he shared his experience with colleagues. By developing techniques to reflect critically on his actions, he realized that he could contribute to making changes towards more appreciation of transition issues at the level of his school.

5 Conclusion

In this paper we describe attempts to elaborate and use a strategy that permits researchers to investigate the calculus transition between high school and university in Tunisia from a continuity perspective. Using this strategy, we paid attention to the cognitive demands of tasks involving conceptual understanding on each side of the transition, to additional examination of this issue through university

Table 8 Classification of teacher and researcher utterances

	Informal statements	Quantifiers rules	Reasoning
Researcher	18	20	28
Teacher	16	38	35
Duration	20 min	35 min	30 min

Table 9 Typical teacher's responses to KQ1 and KQ3

Informal statements	Quantifiers rules	Reasoning
KQ1 "*Is there any problem?*"; "*why do students suggest such properties (of limit)?*"	KQ1 "*I didn't spent enough time speaking about quantifiers rules*"	KQ1 "*Hmm this is a big problem (differences between SC and NC and giving evidence of differences)*"; "*This needs a good counterexample…*"
KQ3 "*n increases and becomes larger…*"; "*should I avoid the using of terms similar to 'approach'…this should help students to comprehend inequalities involved in limit definition*"; "*I have tried to make a link with the graphic (algebraic inequalities)*"	KQ3 "*Students are not expected to know these rules…they are not useful for the following*"; "*It's difficult to find examples that illustrate differences between quantifiers*"; "*How could students understand the need to reverse quantifiers?*"	KQ3 "*I'm not sure that graphics are sufficient…I used to find algebraic examples…*"; "*I have not enough materials (examples of sequences expected by the program)*"

teachers' expectations concerning newcomers' preparation, and to possibilities for high school teachers to reconsider their calculus teaching actions for improving high school students' preparation. On both side of the transition, conceptual understanding of calculus was shaped by typical reasoning, tasks for defining calculus notions, and approximating tasks as tasks that brought out the relationship between the limit notion of Archimedean continuity and real numbers and sequences, differential calculus, and integral calculus. On both sides, calculus definitions were stated for establishing theoretical foundations, and for deductive coherence in the whole course, and calculus ideas were connected through a network of computational knowledge with algebraic inequalities in which limit definitions tied these ideas and techniques together, allowing a few possibilities for the emergence and treatment of students' deep and informal understandings of calculus. For instance, our previous study (Lecorre & Ghedamsi, 2019) on graduate students' conceptions concerning limit shed light on robust conceptions that disallowed them to envisage the use of the limit definition for investigating practical phenomena.

The findings of the complementary study on university teachers' knowledge of the cognitive demands of university tasks and learning needs for facilitating students' transition, show that despite similarities, students enter the university with crucial shortcomings in reasoning and manipulation of formal statements. The application of an experimental collaborative method between mathematics education researchers and teachers shows that it is possible to analyze the extent to which high school teachers can make changes in their teaching actions for the purpose of meeting university teachers' requests. Ideas for future investigations include examination of crucial teachers' calculus knowledge for teaching on both sides of the transition, of the meanings that underlie this knowledge, and of its potential connections.

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